

Generation Interconnection System Impact Study Report

For Queue Project Q17-0856

Elk River 138 kV
September 16, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0856. The Interconnection Customer proposes a 598.5 MW natural gas combined cycle facility interconnecting at the Elk River 138 kV substation in the Pennsylvania area. The request was received on September 01, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 377 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$21,127,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Redondo - Berthusen 138 KV
Contingency: Ontelaunee Energy Ctr - Eliza Crk 345 KV
Loading: 131.0% (Rating: 207 MVA)
DFAX: 5.6%
Pre-project loading: 114.7%
2. Facility: Parkland - Harvey Couch 138 KV
Contingency: Moser Generating Sta - Curecanti 345 KV L
Loading: 114.9% (Rating: 329 MVA)
DFAX: 9.2%
Pre-project loading: 98.2%
3. Facility: Parkland - Ft Smith 138 KV
Contingency: Curecanti - Moser Generating Sta Substation 345 KV Line
Loading: 113.8% (Rating: 343 MVA)
DFAX: 9.2%
Pre-project loading: 97.8%
4. Facility: Kingsburg - Ft Smith 138 KV line
Contingency: Curecanti - Moser Generating Sta 345 KV L
Loading: 112.1% (Rating: 367 MVA)
DFAX: 9.2%
Pre-project loading: 97.1%
5. Facility: Sta Mvalley - Kingsburg Substation 138 KV
Contingency: Curecanti - Moser Generating Station 345 KV
Loading: 110.7% (Rating: 389 MVA)
DFAX: 9.2%
Pre-project loading: 96.6%
6. Facility: Elk Riv - Sta Mvalley 138 KV
Contingency: Curecanti - Moser Generating Station 345 KV L
Loading: 110.0% (Rating: 401 MVA)

DFAX: 9.2%
Pre-project loading: 96.3%

7. Facility: Gray - Leake 138 KV Line
Contingency: Ontelaunee Energy Ctr - Eliza Crk 345 KV
Loading: 108.1% (Rating: 182 MVA)
DFAX: 5.1%
Pre-project loading: 91.2%

8. Facility: Lockeford - Eliza Creek 138 KV
Contingency: Eliza Creek - Ontelaunee Energy Center 345 KV Line
Loading: 104.3% (Rating: 482 MVA)
DFAX: 7.2%
Pre-project loading: 95.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Redondo - Berthusen 138 KV	Rebuild line with higher-capacity conductor	\$3,678,000	55% (\$2,003,000)
Parkland - Harvey Couch 138 KV	Rebuild line with higher-capacity conductor	\$3,914,000	88% (\$3,465,000)
Parkland - Ft Smith 138 KV	Rebuild line with higher-capacity conductor	\$1,275,000	88% (\$1,129,000)
Kingsburg - Ft Smith 138 KV line	Rebuild line with higher-capacity conductor	\$4,292,000	88% (\$3,800,000)
Sta Mvalley - Kingsburg 138 KV line	Rebuild line with higher-capacity conductor	\$3,678,000	88% (\$3,256,000)
Elk Riv - Sta Mvalley 138 KV	Rebuild line with higher-capacity conductor	\$5,977,000	88% (\$5,292,000)
Gray - Leake 138 KV Line	Reconductor line	\$3,108,000	58% (\$1,794,000)
Lockeford - Eliza Creek 138 KV	Reconductor line	\$1,097,000	35% (\$388,000)
	Total allocated to project		\$21,127,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.